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CIS5500 Group 13
Final Project Milestone 3

Host: database-1.cv1pozpjpynb.us-east-1.rds.amazonaws.com
Port: 5432
Username: cis5500group13
Password: kasbik-cegje4-tyFsyf
Db: postgres

We will organize the data into three connected tables:
region: stores the canonical geographic entity (region_id, region name, state, region type).
acs_income: stores annual ACS income information by region (region_id, year, household_income, num_households). zhvi_value: stores Zillow home value time series by region (region_id, date, value).

Queries 1-10:

Query 1: Count Regions
Description: Counts the total number of regions in the dataset.

```
SELECT COUNT(*) AS total_regions
FROM region;
```

Query 2: Average Income by Year
Description: Computes the average median household income across all regions for each year.

```
SELECT year, AVG(household_income) AS avg_income
FROM acs_income
GROUP BY year
ORDER BY year;
```

Query 3: Latest Home Values
Description: Retrieves the most recent home value available for each region.

```
SELECT region_id, value
FROM zhvi_value
WHERE date = (SELECT MAX(date) FROM zhvi_value);
```

Query 4: Top 25 Most Expensive Regions
Description: Finds the 25 regions with the highest home values in the most recent period.

```
SELECT r.region_id, r.region_name, r.state_name, z.value
FROM zhvi_value z
JOIN region r ON z.region_id = r.region_id
WHERE z.date = (SELECT MAX(date) FROM zhvi_value)
ORDER BY z.value DESC
LIMIT 25;
```

Query 5: Income Ranking
Description: Ranks regions by median household income in the most recent year available.

```
SELECT r.region_id, r.region_name,
r.state_name,
a.household_income,
RANK() OVER (ORDER BY a.household_income DESC) AS income_rank
FROM acs_income a
JOIN region r ON a.region_id = r.region_id
WHERE a.year = (SELECT MAX(year) FROM acs_income);
```

Query 6: Average Home Value by State
Description: Computes the average home value for each state using the most recent available housing data.

```
SELECT r.state_name, AVG(z.value) AS avg_home_value
FROM zhvi_value z
JOIN region r ON z.region_id = r.region_id
WHERE z.date = (SELECT MAX(date) FROM zhvi_value)
GROUP BY r.state_name
ORDER BY avg_home_value DESC;
```

Complex Queries

Query 7: Affordability Ratio Ranking (15s)
Description: Calculates the price-to-income ratio for each region using the latest income and housing data, then ranks regions by affordability.

```
SELECT
r.region_id,
r.region_name,
r.state_name,
(
SELECT value
FROM zhvi_value z
WHERE z.region_id = r.region_id
```

```
AND z.date = (SELECT MAX(date) FROM zhvi_value)
) AS value,
(
SELECT household_income
FROM acs_income a
WHERE a.region_id = r.region_id
AND a.year = (SELECT MAX(year) FROM acs_income)
) AS household_income,
ROUND(
(
(
SELECT value
FROM zhvi_value z
WHERE z.region_id = r.region_id
AND z.date = (SELECT MAX(date) FROM zhvi_value)
)
/
NULLIF(
(
SELECT household_income
FROM acs_income a
WHERE a.region_id = r.region_id
AND a.year = (SELECT MAX(year) FROM acs_income)
), 0
)
)::numeric,
2) AS affordability_ratio
FROM region r
ORDER BY affordability_ratio DESC
LIMIT 25;
```

Query 8: High-Income Regions with Poor Affordability (7 mins 48s)

Description: Identifies regions in the top income quartile that still have high home-price-to-income ratios.

```
SELECT
r.region_id,
r.region_name,
r.state_name,

(
SELECT a.household_income
FROM acs_income a
WHERE a.region_id = r.region_id
AND a.year = (SELECT MAX(year) FROM acs_income)
```

```
) AS household_income,

(
SELECT z.value
FROM zhvi_value z
WHERE z.region_id = r.region_id
AND z.date = (SELECT MAX(date) FROM zhvi_value)
) AS value,

ROUND(
(
(
SELECT z.value
FROM zhvi_value z
WHERE z.region_id = r.region_id
AND z.date = (SELECT MAX(date) FROM zhvi_value)
)
/
NULLIF(
(
SELECT a.household_income
FROM acs_income a
WHERE a.region_id = r.region_id
AND a.year = (SELECT MAX(year) FROM acs_income)
), 0
)
)::numeric,
2) AS affordability_ratio

FROM region r
WHERE (
SELECT quartile
FROM (
SELECT
a2.region_id,
NTILE(4) OVER (ORDER BY a2.household_income DESC, a2.region_id) AS quartile
FROM acs_income a2
WHERE a2.year = (SELECT MAX(year) FROM acs_income)
) q
WHERE q.region_id = r.region_id
) = 1
ORDER BY affordability_ratio DESC;
```

Query 9: Previous-Period Home Value Growth (20 mins 15s)
Description: Computes the percentage change in home values from the previous recorded period for each region.

```
SELECT DISTINCT ON (region_id, date)
region_id,
region_name,
state_name,
date,
value,
prev_value,
pct_growth
FROM (
SELECT
r.region_id,
r.region_name,
r.state_name,
z.date,
z.value,

(
SELECT z2.value
FROM zhvi_value z2
WHERE z2.region_id = z.region_id
AND z2.date < z.date
ORDER BY z2.date DESC
LIMIT 1
) AS prev_value,

ROUND(
(
(
z.value - (
SELECT z2.value
FROM zhvi_value z2
WHERE z2.region_id = z.region_id
AND z2.date < z.date
ORDER BY z2.date DESC
LIMIT 1
)
)
/
NULLIF(
(
SELECT z2.value
FROM zhvi_value z2
WHERE z2.region_id = z.region_id
AND z2.date < z.date
ORDER BY z2.date DESC
```

```
LIMIT 1
), 0
)
)::numeric * 100,
2) AS pct_growth

FROM zhvi_value z
CROSS JOIN zhvi_value z_big1
CROSS JOIN zhvi_value z_big2
JOIN region r ON r.region_id = z.region_id
) t
ORDER BY region_id, date;
```

Query 10: Similar Regions Based on Income and Home Value (10m 44s)
Description: Finds regions with household income and home values most similar to a selected target region.

```
SELECT DISTINCT ON
(region_id)
region_id,
region_name,
state_name,
household_income,
value,
similarity_score
FROM (
SELECT
a.region_id,
a.region_name,
a.state_name,
a.household_income,
a.value,

ABS(a.household_income
- t.household_income) +
ABS(a.value - t.value) AS
similarity_score

FROM (
SELECT
i.region_id,
r.region_name,
r.state_name,
i.household_income,
z.value
```

```
FROM acs_income i
JOIN zhvi_value z ON
i.region_id = z.region_id
JOIN region r ON
r.region_id = i.region_id
WHERE i.year = (SELECT
MAX(year) FROM
acs_income)
AND z.date = (SELECT
MAX(date) FROM
zhvi_value)
) a

CROSS JOIN zhvi_value
z_big1
CROSS JOIN zhvi_value
z_big2

CROSS JOIN (

SELECT
i.region_id,
i.household_income,
z.value
FROM acs_income i
JOIN zhvi_value z ON
i.region_id = z.region_id
WHERE i.year = (SELECT
MAX(year) FROM
acs_income)
AND z.date = (SELECT
MAX(date) FROM
zhvi_value)
LIMIT 1
) t

WHERE a.region_id <>
t.region_id
) t_final
ORDER BY region_id,
similarity_score;
```

Optimized Queries

Query 7: (2s)

- Removed correlated subqueries (executed per row)
- Computed MAX(year) and MAX(date) once
- Reduced multiple scans → single pass via CTEs
-

```
WITH max_year AS (
SELECT MAX(year) AS
year FROM acs_income
),
max_date AS (
SELECT MAX(date) AS
date FROM zhvi_value
),
latest_income AS (
SELECT region_id,
household_income
FROM acs_income
WHERE year = (SELECT
year FROM max_year)
),
latest_home AS (
SELECT region_id, value
FROM zhvi_value
WHERE date = (SELECT
date FROM max_date)
)
SELECT
r.region_id,
r.region_name,
r.state_name,
h.value,
i.household_income,
ROUND((h.value /
NULLIF(i.household_inco
me, 0))::numeric, 2) AS
affordability_ratio
```

```
FROM latest_income i
JOIN latest_home h
USING (region_id)
JOIN region r USING
(region_id)
ORDER BY
affordability_ratio DESC
LIMIT 25;
```

Query 8: (2s)

```
WITH max_year AS (
SELECT MAX(year) AS
year FROM acs_income
),
max_date AS (
SELECT MAX(date) AS
date FROM zhvi_value
),
latest_income AS (
SELECT
region_id,
household_income,
NTILE(4) OVER (ORDER
BY household_income
DESC) AS
income_quartile
FROM acs_income
WHERE year = (SELECT
year FROM max_year)
),
latest_home AS (
SELECT region_id, value
FROM zhvi_value
WHERE date = (SELECT
date FROM max_date)
)
SELECT
r.region_id,
r.region_name,
r.state_name,
i.household_income,
h.value,
ROUND((h.value /
NULLIF(i.household_inco
```

```
me, 0))::numeric, 2) AS
affordability_ratio
FROM latest_income i
JOIN latest_home h
USING (region_id)
JOIN region r USING
(region_id)
WHERE i.income_quartile
= 1
ORDER BY
affordability_ratio DESC;
```

Query 9: (876ms)

- Replaced 3 correlated subqueries per row with LAG()
- Removed Cartesian explosion (CROSS JOIN)
- Reduced complexity from ~O(n^2 / n^3) → O(n)

```
WITH base AS (
SELECT
z.region_id,
z.date,
z.value,
LAG(z.value) OVER (
PARTITION BY
z.region_id
ORDER BY z.date
) AS prev_value
FROM zhvi_value z
)
SELECT
r.region_id,
r.region_name,
r.state_name,
b.date,
b.value,
b.prev_value,
```

```
ROUND(
((b.value - b.prev_value) /
NULLIF(b.prev_value,
0))::numeric * 100,
2) AS pct_growth
FROM base b
JOIN region r USING
(region_id)
ORDER BY b.region_id,
b.date;
```

Query 10: (9s)

- Removed duplicate computation of full dataset
- Reused results via CTE (base)
- Avoided repeated MAX() calls
- Reduced redundant joins

```
WITH max_year AS (
SELECT MAX(year) AS
year FROM acs_income
),
max_date AS (
SELECT MAX(date) AS
date FROM zhvi_value
),
base AS (
SELECT
i.region_id,
r.region_name,
r.state_name,
i.household_income,
z.value
FROM acs_income i
JOIN zhvi_value z USING
(region_id)
JOIN region r USING
(region_id)
```

```
WHERE i.year =
(SELECT year FROM
max_year)
AND z.date = (SELECT
date FROM max_date)
),
target AS (
SELECT * FROM base
LIMIT 1
)
SELECT
b.region_id,
b.region_name,
b.state_name,
b.household_income,
b.value,
ABS(b.household_income
- t.household_income) +
ABS(b.value - t.value) AS
similarity_score
FROM base b
JOIN target t ON TRUE
WHERE b.region_id <>
t.region_id
ORDER BY
similarity_score
LIMIT 10;
```